The AI Data-Center Build-Out: Bubble Mechanics, Physical Bottlenecks, And Cost Transmission

A ScienceForge research report on infrastructure constraints and cost transmission

Revision: September 26, 2026 | focused evidence and editorial update

Status: Updated research manuscript; historical model results retained.

Abstract

This report examines how genuine demand for AI infrastructure can coexist with equipment constraints, financing risk and uneven local electricity costs. The September update adds dated primary-source evidence and corrects the interpretation of historical scenario weights. Earlier Forge model classifications remain exploratory; the retained market backtest does not establish selection skill. The useful test is whether operating demand, delivered infrastructure and cost recovery develop together.

September 26 2026 evidence update

Why this matters: More computing can support useful services and scientific work. Building the infrastructure also creates decisions about electricity, land and money. Residents need to know who pays for upgrades; businesses need to know whether promised capacity can actually be delivered.

The IEA's April 2026 assessment reports that global data-center electricity consumption grew 17% in 2025. Its central projection rises from 485 TWh in 2025 to 950 TWh in 2030. The 2030 figure is a projection, not measured consumption. The same assessment explains that efficiency improvements can be offset by wider use and more demanding applications. For local electricity bills, capacity constraints and cost allocation matter: demand growth can raise costs in constrained systems, while spare capacity can change the result.

Virginia provides a concrete example of policy design. The SCC's February 2026 fact sheet describes a separate GS-5 class for qualifying Dominion large-load customers, minimum charges and collateral provisions. New qualifying customers contracting from January 1, 2027 face a 14-year service obligation. These are protections intended to limit cost shifting; their existence is not evidence that every household has already been protected from every increase.

These sources support continued investigation of physical constraints and regional cost allocation. They do not validate the Forge's historical market scores, ticker groups, national shock label or crash probabilities.

Questions readers can use

- **For a proposed local project:** How much demand is contracted, how much is operating, and who covers unused capacity if the project is delayed or cancelled?
- **For electricity bills:** Which costs are attributed to this project, which are shared system costs, and what do the approved tariff and contract actually recover?
- **For the build-out thesis:** Are connections and equipment arriving faster, and are useful workloads supporting the operating costs?
- **For the next research update:** Compare actual load, project completions and rate outcomes against dated expectations. Keep requests for grid capacity separate from operating demand.

Corrections to the interpretation of earlier results

The older scenario weights total 1.0, but their different time horizons and potentially overlapping outcomes do not establish a mutually exclusive probability distribution. This revision retains them only as historical research weights. A future probability forecast needs a common dated horizon, non-overlapping resolution rules and outcome scoring. No weights have been recalibrated here.

Relative timelines such as “now to two quarters” refer to the original model run, whose precise anchor is not established in this manuscript. They must not be restarted on September 26. Dated shipping advisories and source-registry labels remain historical evidence unless individually refreshed. Source registration is not the same as verification of a current claim.

Retained research analysis and historical model results

Reading the following sections: Model scores, backtests, source statuses, historical validation records below are carried forward from the original manuscript. They were not rerun or independently revalidated in this editorial update. The original source snapshot is retained privately; this public report includes the historical claims and source references.

Executive Summary

This paper frames the AI data-center boom as a researchable economic regime rather than a simple prediction call. The recorded Forge read is strain_without_break with bubble phase reflexive_buildout. The strongest version of the thesis is that demand remains real while physical bottlenecks and financing/crowding risks create a conditional bubble dynamic. The recorded 21-window survivor/fragility test has a 0.57 hit rate and 4.20 percentage-point average spread. This is null-to-weak evidence, not demonstrated selection skill.

Plain English: The question is not whether AI demand is fake. The question is whether real demand is running into power, parts, financing, and regional cost limits faster than markets expect.

Technical Read: Treat the build-out as a bottleneck-transmission regime: scarcity rents can coexist with fragile financing, regional ratepayer pressure, project delays, and valuation overshoot.

Research Question

Can the data-center build-out be explained by a bottleneck-transmission model that predicts which industries retain pricing power, which regions see power-cost pressure, and when bubble-like financial fragility overtakes genuine infrastructure demand?

Plain English: We want to separate the winners, the squeezed suppliers, the exposed regions, and the weak players before the market makes that split obvious.

Technical Read: The target model should estimate component bottleneck persistence, regional rate/capex transmission, financing fragility, and scenario transitions without relying on one deterministic burst date.

Evidence Base And Current Read

The data-center watch packet reports: Current Forge read: strain without break. Physical strain is high at 0.94, crowding is 0.58, and support is 0.85. That argues for a crowded and fragile build-out, but not yet a clean burst. The build-out map reports stage compute_bound and macro stress 0.4046. Top component constraints: HBM, DRAM, NAND, and enterprise storage (critical, bottleneck 0.874, catch-up 0.3238); AI accelerators and server compute (critical, bottleneck 0.866, catch-up 0.3261); Networking, switching, optics, and interconnect (critical, bottleneck 0.8599, catch-up 0.3325); Cooling, liquid cooling, thermal systems, and mechanical infrastructure (critical, bottleneck 0.821, catch-up 0.2958); Copper, aluminum, land, fiber, and data-center real estate (critical, bottleneck 0.7965, catch-up 0.3792).

Plain English: The Forge has useful signals, but this is still an evidence-building draft. It has enough to guide investigation, not enough to declare proof.

Technical Read: The current evidence stack combines Forge watch packets, power-crunch packets, local EIA archive metadata, and measured EIA retail-rate cache evidence. Utility filings and RTO/ISO evidence are now joined to the regional

exposure model across multiple supporting regions. This closes the mechanical replication gate, but does not by itself establish national causality or eliminate confounding. The utility/RTO index has discovered 13 candidate filing and grid-operator sources across 6 regions, with 7 parsed records; multi-region support is available for 3 regions, while national causal interpretation remains bounded by comparison-region and outcome limits.

Bubble Mechanics Framework

The paper separates bubble formation from burst timing. Demand can stay strong while a bubble forms if investors extrapolate scarcity profits, if second-tier participation broadens, or if financing quality degrades. Timestamped exploratory model-state outputs are: formation 0.73, psychological heat 0.57, capex-reality gap 0.67, financing fragility 0.62.

Plain English: Bubbles usually grow while the story still sounds right. The danger point is when expectations outrun cash flow, physical capacity, or financing quality.

Technical Read: The framework separates formation heat, psychological participation, capex-reality gap, financing fragility, survivor quality, and weak-player exposure. That keeps the paper from treating a broad correction and a total demand collapse as the same event.

Market Psychology, Pullbacks, And Panic Transmission

The paper treats psychology as a research hypothesis about a possible transmission layer. In infrastructure bubbles, a pullback may start before the physical story is disproved: investors first stop assuming every participant wins, then financing and liquidity may determine whether the move remains a valuation reset or becomes a panic. The recorded Forge psychology read is valuation_reset_risk. The heat and panic-transmission scores are timestamped exploratory model-state outputs, not validated thresholds or causal laws. The new breadth artifact covers 10 symbols, with 50-day survivor-minus-fragility breadth spread -0.2 and 200-day spread 0.4. The behavioral-leverage cascade is currently leverage_reflexivity_watch with score 0.5262. Panic confirmation is false; the rule requires dated credit/liquidity evidence plus multiple market-structure confirmations. The issuer leverage panel screens 16 of 16 watchlist issuers. 1 show funding dependence in the current evidence, while 0 have issuer-specific primary evidence of refinancing or forced-selling pressure (tickers: none). The historical signature comparison is mixed_signature. Its leading mechanism match is Ordinary crowded-sector correction (similarity 0.77), followed by 2000-style early valuation and leadership unwind (similarity 0.75). These are feature comparisons, not event probabilities or claims that history will repeat.

Plain English: Markets can turn while demand still looks strong. The first warning is usually a change in belief: investors start asking who actually earns returns, who is overbuilding, and who needs funding. Panic risk rises when that doubt collides with weak credit or forced selling.

Technical Read: The psychology layer combines narrative heat, capex-reality gap, support deterioration, financing fragility, power/capex pressure, and breadth quality. The behavioral cascade then tests whether incentives and leverage have progressed from speculative expansion into refinancing stress and forced selling. The issuer panel separates high spending from actual funding dependence and requires dated issuer evidence before escalating to refinancing or forced-selling risk. It separates a valuation reset from a panic/liquidity break, which matters because the right evidence and timing signals are different.

Historical Market Pattern Comparison

The strongest current mechanism match is Ordinary crowded-sector correction, with 2000-style early valuation and leadership unwind also present. This is a signature comparison, not a claim that the historical episode will repeat. The comparison explicitly separates an equity and expectations unwind from credit transmission, market-structure liquidation, a rate-led growth repricing, and an ordinary crowded-sector correction. Current ranked comparison: Ordinary crowded-sector correction: similarity 0.77. Critical distinction: This remains the benign control case and requires visible confidence repair, not merely calm credit. 2000-style early valuation and leadership unwind: similarity 0.75. Critical distinction: The early dot-com unwind was initially an equity and expectations break; full-cycle similarity would require a prolonged earnings and business-model reset. 2022-style growth-sector valuation reset: similarity 0.72. Critical distinction: This pattern allows a

severe repricing while the central technology and stronger businesses continue expanding. 1987-style market-structure and forced-liquidation shock: similarity 0.59. Critical distinction: This signature emphasizes market plumbing and speed rather than a long deterioration in underlying economics. 2007-style credit and balance-sheet transmission: similarity 0.09. Critical distinction: A 2007-style state is fundamentally a credit-transmission event; falling technology shares without funding-market stress are insufficient.

Plain English: The market does not have to repeat 2000, 2007, or 1987 exactly. The useful question is which mechanism is operating now: investors rejecting extreme expectations, lenders withdrawing funding, forced sellers overwhelming buyers, or a temporary correction repairing itself.

Technical Read: The matcher uses current correction pressure, sector concentration, breadth, failed rallies, downside volume, benchmark contagion, fresh credit breaches, issuer funding pressure, and support-break evidence. Every historical label retains its contradictions and next confirmations.

Forge Timeline Estimate

The recorded Forge estimate rejects a single hard crash date. The older late-Q3 2026 hard-burst call now looks too precise. The updated stack still sees real physical strain, but revenue and revision support remain strong enough that the cleaner read is strain and crowding first, break later only if support rolls over. The structured estimate is: Now to 2 quarters: strain_without_break (Build-out remains active and physically strained. Current support/revisions are strong enough that a clean burst is not the base case.) 2 to 4 quarters: crack-watch transition if support deteriorates (This is the first serious window for visible market stress: second-tier rollover, funding selectivity, and capex discipline can appear before demand itself fails.) 12 to 36 months: regional cost crunch base case (Power, interconnect, transformer, cooling, and memory constraints are most likely to show up as regional cost pass-through and project triage.)

Plain English: The useful forecast is a watch list, not a prophecy. Risk rises or falls as power queues, equipment lead times, rates, revisions, and financing data move.

Technical Read: Use conditional scenario windows and trigger thresholds rather than one exact burst date. The timeline should be recalibrated as realized rate pressure, project delays, capex revisions, and bottleneck catch-up indicators arrive.

Power, Supply Chain, And Cost Transmission

The power-crunch packet reads national stage shock with power-crunch score 0.7841, ratepayer-pressure score 0.7766, and project-delay score 0.7984. This supports a cost-transmission research lane: power, grid equipment, memory, cooling, copper, land, and labor can raise costs even before end-demand fails. The magnitude and incidence depend on tariff design, cost-allocation rules, customer protections, and whether new generation and transmission arrive before load. The current EIA retail-rate cache adds measured residential price evidence for 13 exposed states with average rate-pressure score 0.3887 and max score 0.7141; this improves measurement with multi-region utility/RTO support now present, while national causal interpretation remains bounded by comparison-region and outcome limits. The utility/RTO index has discovered 13 candidate filing and grid-operator sources across 6 regions, with 7 parsed records; multi-region support is available for 3 regions, while national causal interpretation remains bounded by comparison-region and outcome limits.

Plain English: Electric bills and project costs can rise even if households use the same power, because grid upgrades and capacity costs can be spread across customers.

Technical Read: Ratepayer transmission needs triangulation across EIA retail rates, utility rate cases, tariff riders, RTO/ISO interconnection queues, capacity markets, and local capex plans before the paper can claim data-center causality. Regulatory cost-shielding provisions in states such as Georgia, Texas, and Virginia must be modeled as potential offsets rather than assumed away.

Financial Deflation And Margin-Normalization Risk

The paper separates physical scarcity from financial deflation. The recorded Forge financial-deflation lane is caution with aggregate score 0.4869 and top lane Electricity, capacity prices, and firm power contracts. This means a supplier or

component can remain physically important while its scarcity premium, margin, or valuation multiple normalizes.

Plain English: A bubble does not have to end because demand disappears. It can also cool because supply catches up, margins compress, or investors stop paying for every scarce input as if scarcity will last forever.

Technical Read: The financial-deflation lane models pricing-power fade, crowding reset, market risk, support deterioration, and catch-up pressure separately from physical bottlenecks. It remains modeled until measured ASP, inventory, backlog, and margin data are added.

Geopolitics, Shipping, And Energy Chokepoints

The paper now treats geopolitics as a separate transmission lane rather than a generic risk note. The recorded Forge geopolitical-shipping lane is caution with aggregate score 0.6103 and top chokepoint Red Sea / Suez routing risk. Gulf, Red Sea, Suez, Panama, and other chokepoint disruptions can change delivery timing, freight rates, marine insurance, oil/LNG risk premiums, and investor psychology before they change real AI infrastructure demand.

Plain English: A shipping or Gulf energy shock does not have to destroy demand to matter. It can make the build-out slower, more expensive, and more fragile.

Technical Read: Model chokepoint stress as a logistics, energy, insurance, and risk-premium factor. Keep it separate from physical power constraints and market psychology, then join it through scenario branches and evidence gates after live route-flow, freight, insurance, energy, and company-delay data are refreshed.

Survivors, Fragility, And Market Structure

The preliminary survivor basket includes MU, WDC, AMD, AMAT, LRCX. The preliminary fragility watch includes KLAC, SMH, SMCI, DELL, VRT. These labels are provisional research groups, not predictions of survival or failure and not investment recommendations. The narrower falsifiable hypothesis is that bottleneck-exposed firms may separate from capacity-expanding or crowded participants over time if pricing power, revisions, funding, and relative returns support that distinction. The survivor/fragility backtest has 21 rolling 60-day windows, hit rate 0.5714, and average spread 4.202 percentage points. The current result is null-to-weak rather than proof of predictive skill, and remains vulnerable to in-sample selection and limited-cycle risk.

Plain English: Some firms can still win after a bubble cools. The weaker or crowded players are usually the ones that get repriced first.

Technical Read: The bottleneck-exposed versus capacity-expanding split should be measured through margins, revisions, balance-sheet strength, pricing power, valuation compression, and relative returns against a benchmark, with constituents fixed before each evaluation window and tested out of sample.

Scenario And Falsification Plan

The outcome distribution is recorded in the earlier model: Base Case: Regional Cost Crunch at historical research weight 0.52 over 12-36 months: Regional power markets absorb data-center load through higher utility capex, capacity costs, and firm-power contracting. Household bills may rise in exposed regions even if usage is flat, depending on supply and cost allocation. Upside: Supply And Grid Catch-Up at historical research weight 0.24 over 18-48 months: Grid equipment supply improves, interconnect queues shorten, storage and demand response expand, and memory/accelerator supply catches up. Project delays ease and rate pressure becomes more manageable. Downside: Rate Shock And Project Triage at historical research weight 0.24 over 6-24 months: Large-load growth outruns local grid upgrades. Utilities and grid operators force project triage, power contract premiums rise, and ratepayer pressure becomes politically salient. The earlier arithmetic audit recorded a sum of 1.0. That does not establish mutually exclusive outcomes or calibrated probabilities; the September correction above governs interpretation. Overlay risks may overlap these scenarios. The base scenario is sustained demand with intermittent cost/pass-through stress. The downside scenario is second-tier rollover followed by capex discipline. The upside scenario is a supply response that relieves bottlenecks without collapsing demand. Falsifiers include normalized interconnect queues, falling equipment lead times, improving utility rate pressure in exposed regions, no performance split

between survivor and fragility baskets, capex-guidance changes, GPU rental-price changes, and chokepoint/energy shock data.

Plain English: The paper should be able to lose. If supply catches up, rates calm down, and weak players do not underperform, the thesis has to be narrowed or rejected.

Technical Read: Pre-register branch probabilities, falsifiers, evidence thresholds, and maturity dates. Score the result later through the prediction/calibration ledger; overlay probabilities must be calibrated separately from the mutually exclusive base outcomes.

Remaining Validation:

- Score matured scenario outcomes before making calibration claims.
- Wait for scenario deadlines to mature, then record outcomes in the Prediction OS ledger.

Limitations And Evidence Standards

The retained analysis covers market psychology, utility evidence, equipment supply, shipping risks and backtests. The remaining scientific work includes scoring dated scenario outcomes and testing falsifiers against actual observations.

Plain English: The paper is complete as an evidence-bounded internal research report. It is not a claim that a crash will happen or a specific-date prediction. The Forge will keep updating it as the scenarios mature.

Technical Read: Explanatory figures are not empirical evidence. Forecast validation depends on observed outcomes and scoring; uncertainty labels and conditional timelines remain necessary.

Registered Timeline And Watch Triggers

- **Now to 2 quarters** - strain_without_break | likelihood 0.71 | confidence 0.58: Build-out remains active and physically strained. Current support/revisions are strong enough that a clean burst is not the base case.
- Watch trigger: estimate revisions roll over
- Watch trigger: hyperscaler utilization/capex language weakens
- Watch trigger: project delays become margin pressure rather than scarcity pricing
- **2 to 4 quarters** - crack-watch transition if support deteriorates | likelihood 0.47 | confidence 0.52: This is the first serious window for visible market stress: second-tier rollover, funding selectivity, and capex discipline can appear before demand itself fails.
- Watch trigger: weaker second-tier equity/credit divergence
- Watch trigger: large customer capex discipline
- Watch trigger: HBM or accelerator pricing momentum fades while inventories rise
- **12 to 36 months** - regional cost crunch base case | likelihood 0.52 | confidence 0.55: Power, interconnect, transformer, cooling, and memory constraints are most likely to show up as regional cost pass-through and project triage.
- Watch trigger: utility rate-case acceleration
- Watch trigger: RTO/ISO interconnect queue persistence
- Watch trigger: equipment backlog-to-revenue conversion stalls
- **18 to 48 months** - supply catch-up or margin normalization | likelihood 0.24 | confidence 0.46: If supply catches up, the bubble may resolve through pricing/margin normalization rather than a demand crash.
- Watch trigger: falling power-contract premiums
- Watch trigger: inventory days normalize without demand destruction

- Watch trigger: catch-up indicators improve for two consecutive reports

Scenario Families

Historical scenario weights

These preserved research weights sum to 100% arithmetically. The differing horizons and overlapping mechanisms prevent interpreting them as a validated outcome distribution.

- **Base Case: Regional Cost Crunch** (12-36 months, historical research weight 0.52): Regional power markets absorb data-center load through higher utility capex, capacity costs, and firm-power contracting. Household bills may rise in exposed regions even if usage is flat, depending on supply and cost allocation.
- Bubble overlay: Forge overlay: bubble formation is elevated, but the current strain read demotes an exact hard-burst date to a watch condition.
- **Upside: Supply And Grid Catch-Up** (18-48 months, historical research weight 0.24): Grid equipment supply improves, interconnect queues shorten, storage and demand response expand, and memory/accelerator supply catches up. Project delays ease and rate pressure becomes more manageable.
- Bubble overlay: Forge overlay: catch-up resolves pressure through normalization, not a crash.
- **Downside: Rate Shock And Project Triage** (6-24 months, historical research weight 0.24): Large-load growth outruns local grid upgrades. Utilities and grid operators force project triage, power contract premiums rise, and ratepayer pressure becomes politically salient.
- Bubble overlay: Forge overlay: bubble formation is elevated, but the current strain read demotes an exact hard-burst date to a watch condition.

Non-Exclusive Overlay Risks

These risks may overlap with any base scenario, so their weights do not add to 100%.

- **Financial Deflation Without Demand Collapse** (6-24 months, research weight 0.34): Physical build-out demand remains real, but scarcity margins, valuations, or second-tier supplier expectations normalize as supply catches up or investors stop extrapolating bottleneck profits.
- **Geopolitical Shipping And Energy Shock** (event-triggered 0-18 months, research weight 0.31): Gulf, Red Sea, Suez, or other chokepoint stress raises logistics, insurance, oil/LNG, and delivery-delay pressure. The build-out can remain demand-supported while project timing, component buffers, and energy costs worsen.

Trigger-Gated Watch Conditions

Activation scores are internal monitoring indices, not standalone probabilities.

- **Psychology-Led Pullback: Narrative Reset** (event-triggered 1-12 months, activation score 0.47): A pullback can begin before physical demand fails if the market psychology shifts from endless capacity scarcity to ROI scrutiny, utilization discipline, financing selectivity, and second-tier rollover.
- **Hard Burst Tail: Financing Break Or Demand Repricing** (conditional 6-24 months, activation score 0.5): A hard bubble break is not the current base case; it becomes credible if revisions, utilization, financing quality, and breadth all deteriorate together.

Historical figure specifications

- **Forge timeline estimate for the AI data-center build-out** timeline / renderable_review_visual - Conditional timeline based on current Forge scores. It is not a deterministic crash-date claim.

- **Build-out pressure scorecard** score_bars / renderable_review_visual - Score bars summarize the Forge packet; the evidence matrix remains the source of truth.
- **Top physical bottlenecks and catch-up scores** component_bars / renderable_review_visual - High bottleneck and low catch-up scores imply scarcity/cost pressure can persist even while demand is real.
- **Historical scenario weights** outcome_bars / renderable_review_visual - These exploratory weights are not calibrated event probabilities; the September correction governs interpretation. No historical chart is reproduced here.

Claims And Falsifiers

- **SF-ECON-CLAIM-B07DD01003B9** The AI data-center cycle is best modeled as a bottleneck-transmission regime: the tightest physical layer currently led by HBM, DRAM, NAND, and enterprise storage is hypothesized to influence pricing power, project delays, and industry outcomes alongside raw demand. Falsifier: The claim weakens if component bottleneck scores normalize while project delays, pricing power, and revision dispersion remain elevated.
- **SF-ECON-CLAIM-AD1DB7C724CF** Current bubble risk is conditional rather than deterministic: the Forge reads the stage as strain_without_break and the bubble phase as reflexive_buildout, so the research question is not simply whether demand is real, but when support, financing quality, and physical constraints diverge. Falsifier: The claim weakens if bubble formation, crowding, and financing fragility remain low for consecutive reports despite sustained capex.
- **SF-ECON-CLAIM-827BA7ACCF7D** Data-center load growth can transmit into household and business costs through capacity prices, transmission upgrades, utility capital recovery, and firm-power contracts; the magnitude and incidence depend on tariff design, cost-allocation rules, customer protections, and whether new generation and transmission arrive before load. Falsifier: The claim weakens if regional retail-rate evidence and utility filings show no relationship between high data-center exposure and rate/capex pressure.
- **SF-ECON-CLAIM-2048E7E7A107** The Forge currently classifies the national power-crunch stage as shock; this is a timestamped model-state assertion, not an independently measured national causal finding. Falsifier: The model-state assertion changes when the next source refresh moves the declared power-crunch stage out of shock.
- **SF-ECON-CLAIM-C8CD1DCD6CF0** Bubble mechanics should separate provisional survivor/fragility research groups by bottleneck exposure, capacity expansion, and valuation risk: firms with durable pricing power, favorable revisions, and balance-sheet support may still outperform even if weaker or more crowded second-tier names are later repriced. Falsifier: The claim weakens if survivor candidates and fragility candidates do not separate in margins, revisions, or relative returns over the backtest horizon.
- **SF-ECON-CLAIM-897637AF7ACD** Market psychology is an independent transmission layer in the data-center cycle: a pullback can start from a narrative reset before demand fails, while panic risk may become more material when measured psychological heat combines with financing fragility, breadth deterioration, and liquidity stress; the recorded Forge correction style is valuation_reset_risk. The heat score and panic threshold are exploratory model-state outputs, not established causal thresholds. Falsifier: The claim weakens if psychological heat stays elevated but pullbacks remain shallow, breadth improves, credit remains loose, and guidance continues to broaden rather than concentrate.
- **SF-ECON-CLAIM-A9C4E467E8F5** Geopolitical chokepoint stress can affect the AI data-center build-out through shipping delays, marine insurance, energy prices, and risk premiums before it changes real demand; the recorded Forge geopolitical-shipping lane is caution with top chokepoint Red Sea / Suez routing risk. The original shipping-risk engine status was fresh_shipping_and_geopolitical_risk; it is not a September 26 freshness finding. The original run reported with 2 risk events and 2 official geopolitical assessments loaded. Falsifier: The claim weakens if chokepoint flow data, shipping rates, war-risk insurance, oil/LNG prices, and company delivery language remain calm during the proposed geopolitical stress window.

Evidence Matrix

- **SF-ECON-CLAIM-B07DD01003B9** status derived_packet_supported_hypothesis, confidence 0.68, receipts: SF-ECON-RCPT-9AA81AAAE52D, SF-ECON-RCPT-291CACC66102, SF-ECON-RCPT-8681EEA6FF9C,

SF-ECON-RCPT-EBDEBC6F2D21, SF-ECON-LEAF-B80276CCFD4A, SF-ECON-LEAF-455E79778F9E, SF-ECON-LEAF-A0147824C8A6, SF-ECON-LEAF-964C88978680, SF-ECON-LEAF-3B3E3A348131, SF-ECON-LEAF-D16B3C38A891, SF-ECON-LEAF-DF3792D87AC1, SF-ECON-LEAF-30065FEBF040, SF-ECON-LEAF-885AB592F4FE

- Promotion blocker: needs_company_margin_sensitivity

- **SF-ECON-CLAIM-AD1DB7C724CF** status derived_packet_supported_hypothesis, confidence 0.54, receipts: SF-ECON-RCPT-9AA81AAAE52D, SF-ECON-RCPT-291CACC66102, SF-ECON-LEAF-5303581C067B, SF-ECON-LEAF-1CDF429528E2, SF-ECON-LEAF-5CEE77A2C6A6

- Promotion blocker: needs_pre_registered_burst_trigger_model

- Promotion blocker: needs_bubble_score_history

- Promotion blocker: needs_second_tier_financing_evidence

- **SF-ECON-CLAIM-827BA7ACCF7D** status derived_packet_supported_hypothesis, confidence 0.61, receipts: SF-ECON-RCPT-1218F8AB2C7A, SF-ECON-RCPT-EA8858A5D74E, SF-ECON-RCPT-1392EE1D4222, SF-ECON-LEAF-EA8D22923777, SF-ECON-LEAF-A912ED6FF535, SF-ECON-LEAF-5BEBDF51F0E2, SF-ECON-LEAF-8D2A1131BCD0, SF-ECON-LEAF-53DD356FFD70, SF-ECON-LEAF-13D08FA1E3C6, SF-ECON-LEAF-2D6446203674, SF-ECON-LEAF-31604FD300DB, SF-ECON-LEAF-D86B6A6A5825, SF-ECON-LEAF-B9486183FBB5, SF-ECON-LEAF-B80276CCFD4A

- **SF-ECON-CLAIM-2048E7E7A107** status forge_model_state_only, confidence 0.42, receipts: SF-ECON-RCPT-1218F8AB2C7A

- Promotion blocker: internal_model_state_not_external_empirical_claim

- **SF-ECON-CLAIM-C8CD1DCD6CF0** status derived_packet_supported_hypothesis, confidence 0.55, receipts: SF-ECON-RCPT-291CACC66102, SF-ECON-RCPT-6D0086048C87, SF-ECON-LEAF-5CEE77A2C6A6

- Promotion blocker: needs_balance_sheet_funding_evidence

- **SF-ECON-CLAIM-897637AF7ACD** status derived_packet_supported_hypothesis, confidence 0.58, receipts: SF-ECON-RCPT-9AA81AAAE52D, SF-ECON-RCPT-291CACC66102, SF-ECON-RCPT-6D0086048C87, SF-ECON-RCPT-839B5F3EEB9C, SF-ECON-LEAF-E561CE96A1D6

- **SF-ECON-CLAIM-A9C4E467E8F5** status shipping_engine_and_official_geopolitical_context_present_needs_market_confirmation, confidence 0.43, receipts: SF-ECON-RCPT-932D4B434E32, SF-ECON-RCPT-D192DDB78460, SF-ECON-RCPT-DB73BF917E49, SF-ECON-RCPT-CDE751F87A9F, SF-ECON-RCPT-48088EE445D5, SF-ECON-RCPT-2A1AE77B08DB, SF-ECON-LEAF-D9125BD30499, SF-ECON-LEAF-E7AD00CF08AA

Source Health And Receipts

- **Primary source pack**: present_needs_claim_review; present 7/7 required sources

- Limitation: The manifest closes the source-pack organization blocker, not the proof or causality gates.

- Limitation: Tracking and candidate-index lanes still require exact source support before public claims.

- Limitation: Replicated transformer, memory/HBM, geopolitical, market-psychology, and scenario-outcome calibration remain separate gates.

- **forge_data_center_watch**: ok

- **forge_power_crunch**: ok

- **eia_retail_rate_cache**: ok

- **eia_local_archives**: cached

- **utility_rto_evidence_index**: supporting
- **transformer_supply_evidence**: complete
- **memory_chip_supply_evidence**: complete
- **score_provenance**: ok
- **geopolitical_shipping**: shipping_engine_official_geo_and_live_market_context_present
- **live_geopolitical_shipping_evidence**: present
- **forge_shipping_risk_engine**: fresh_shipping_and_geopolitical_risk
- SF-ECON-RCPT-9AA81AAAE52D Data center bottleneck transmission and bubble mechanics packet - Stage strain_without_break with physical strain 0.9375, bubble formation 0.7348, phase reflexive_buildout. Component stack: HBM, DRAM, NAND, and enterprise storage bottleneck 0.874 severity critical; AI accelerators and server compute bottleneck 0.866 severity critical; Networking, switching, optics, and interconnect bottleneck 0.8599 severity critical; Cooling, liquid cooling, thermal systems, and mechanical infrastructure bottleneck 0.821 severity critical; Copper, aluminum, land, fiber, and data-center real estate bottleneck 0.7965 severity critical. Verdict: Current Forge read: strain without break. Physical strain is high at 0.94, crowding is 0.58, and support is 0.85. That argues for a crowded and fragile build-out, but not yet a clean burst.
- SF-ECON-RCPT-291CACC66102 Psychological heat, capex-reality gap, survivor, and fragility scores - Bubble phase reflexive_buildout with psychological heat 0.5747, capex reality gap 0.667, financing fragility 0.6196, survivor quality 0.6666, fragility 0.5281.
- SF-ECON-RCPT-1218F8AB2C7A Regional power and ratepayer-pressure packet - National stage shock with power crunch score 0.7841, ratepayer pressure 0.7766, and project delay score 0.7984. Plain-English read: The Forge read is not a clean national blackout call. It is a regional cost-and-delay crunch: new data-center and industrial load forces grid upgrades, firm power contracts, and equipment procurement. Those costs can reach households through retail rates, rid...
- SF-ECON-RCPT-5A2EE3958768 Local EIA projection and balancing-authority source context - Indexed 3 local archives (AEO2025.zip, AEO2026.zip, EBA.zip) with lanes energy_price_projection, generation_capex_projection, hourly_generation, natural_gas_context, operational_grid_context, projection_context, storage_cost_or_operation. AEO series provide projection/model context; EBA series provide operational balancing-authority context. This receipt does not substitute for measured retail-rate evidence.
- SF-ECON-RCPT-EA8858A5D74E Measured state-level residential retail electricity-rate evidence - Cached EIA retail-sales evidence covers 13 exposed states. Average rate-pressure score 0.3887; maximum rate-pressure score 0.7141. Multi-region utility/RTO support is now present; national causal interpretation remains bounded by comparison-region and outcome limits.
- SF-ECON-RCPT-1392EE1D4222 Indexed utility, regulator, and grid-operator source candidates for regional cost-transmission evidence - Indexed 13 utility/RTO candidate sources across 6 regions, with 7 parsed records and 3 supporting regions carrying both filing and operator support. This supports multi-region cost-transmission review when parsed support exists; national causal interpretation remains bounded by comparison-region and outcome limits.
- SF-ECON-RCPT-8681EEA6FF9C Source-indexed transformer/switchgear lead-time, backlog, and supplier-filing evidence lane - Transformer/switchgear evidence status complete with 6 source candidates, 6 source records, 3 parsed lead-time records, 3 parsed backlog records, and 3 supplier records. Blockers: none. This narrows the grid-equipment evidence search but does not prove current supply tightness until dated numeric records are parsed.
- SF-ECON-RCPT-EBDEBC6F2D21 Source-indexed RAM/HBM, capex/capacity, and advanced-packaging evidence lane - Memory/chip evidence status complete with 7 source candidates, 7 source records, 3 HBM/DRAM records, 2 capex/capacity records, and 2 advanced-packaging records; 12 validated primary-source receipts across 3 suppliers. Blockers: none. This narrows the RAM crisis and chip-production evidence search while still requiring numeric pricing/capacity calibration.

- SF-ECON-RCPT-7F9594B0A450 Modeled market-structure lane for pricing-power fade and margin normalization - Financial-deflation stage caution with aggregate score 0.4869 and top lane Electricity, capacity prices, and firm power contracts. This lane is modeled from component catch-up, crowding, market risk, and support; it is not measured ASP evidence.

- SF-ECON-RCPT-D7BFED6EF1C2 Modeled cost-stack sensitivity for energy, memory, grid, cooling, and construction pressure - Baseline modeled cost pressure 0.7732, cost-escalation risk 0.7456, economic headroom 0.5542, and current read viable_but_margin_sensitive. The +10%, +25%, and +50% scenarios test whether rising input costs can force selective project deferrals before demand destruction. This is a normalized model aid, not a dollar project-return estimate.

- SF-ECON-RCPT-932D4B434E32 Modeled chokepoint, logistics, insurance, and energy-risk transmission lane - Geopolitical-shipping stage caution with aggregate score 0.6103 and top chokepoint Red Sea / Suez routing risk. This lane is source-context until refreshed with live PortWatch, shipping-rate, war-risk insurance, energy-price, and company delivery-delay evidence.

- SF-ECON-RCPT-D192DDB78460 Latest Forge shipping-risk report linked into the power-crunch and paper evidence stack - Shipping-risk status fresh_shipping_and_geopolitical_risk from report the retained internal shipping-risk receipt dated July 8, 2026; risk events 2, shipping-risk score 1.0, official geopolitical assessments 2. Top chokepoints: Lombok Strait, Malacca Strait, Suez Canal, Bab-el-Mandeb, Panama Canal, Strait of Hormuz. Exposed commodities: copper, iron_ore, crude_oil, lng. This is internal engine evidence and does not replace PortWatch, insurance, energy, or company-delay receipts.

- SF-ECON-RCPT-DB73BF917E49 Live public-source corroboration lanes for route-flow, freight, insurance, energy, and delivery-delay context - Live source-context status present with 5/5 required lanes present. Lane statuses: portwatch_route_flow=source_available; shipping_rate_context=source_available; war_risk_insurance_context=source_reachable_restricted; eia_energy_series=series_available; company_delivery_delay_context=source_reachable_restricted. Missing lanes: none. This supports live corroboration/source-health only; causal promotion still needs dated event extraction and company delivery-delay excerpts.

- SF-ECON-RCPT-CDE751F87A9F Dated geopolitical event records and company delivery-delay records for shipping/energy scenario review - Causal-extraction status complete with 2 dated event records and 2 company delivery-delay records. Blockers: none. This closes dated extraction only; quantitative realized-impact calibration remains required.

- SF-ECON-RCPT-48088EE445D5 Source-linked freight, insurance, energy, and delivery-delay impact-lane calibration - Impact calibration status complete with 4/4 impact lanes present, 2 event records, and 2 company delay records. Blockers: none. This closes source-linked impact-lane coverage only; numeric impact estimates remain a separate evidentiary question.

- SF-ECON-RCPT-2A1AE77B08DB Numeric-delta and proxy record layer for freight, insurance, energy, and delivery-delay lanes - Numeric-delta replication status complete with 4 records across 4/4 lanes. Lane deltas: shipping_rate=queue_growth_pct_proxy:35.0percent; insurance=active_war_risk_advisory_count:2active_advisories; energy_price=RNGWHHD_percent_delta:8.9404$/MMBTU; delivery_delay=max_eta_or_dwell_delay_hours_proxy:72.0hours. Blockers: none. This attaches bounded numbers to the geopolitical lane; proxy limitations still block over-strong causal magnitude claims.

- SF-ECON-RCPT-6D0086048C87 Measured market breadth and survivor-versus-fragility tracking artifact - Breadth status pass with coverage 10, 50-day breadth spread -0.2, and 200-day breadth spread 0.4. Backtest status pass with 21 rolling windows, hit rate 0.5714, average spread 4.202 pct, latest spread 6.3313 pct. This starts the measured market-structure lane but does not replace risk-adjusted replication, credit/liquidity evidence, or transcript evidence.

- SF-ECON-RCPT-839B5F3EEB9C Narrative, credit/liquidity, and company guidance-language source context for the data-center bubble paper - Market-psychology evidence status complete with 2 narrative records, 3 credit/liquidity records, and 2 guidance-language proxy records. Blockers: none. This narrows the psychology gate but does not replace transcript-corpus calibration or realized outcome scoring.

- SF-ECON-LEAF-EA8D22923777 doe-lpt-resilience-2024-supply-chain-vulnerability - DOE report frames large power transformer supply as a resilience and supply-chain vulnerability.
- SF-ECON-LEAF-A912ED6FF535 doe-trac-lpt-lead-time-lower-bound - Large power transformers are custom engineered and can require lead times of one year or more.
- SF-ECON-LEAF-5BEBDF51F0E2 ERCOT batch connection process for large electricity users - ERCOT says large-user connection requests, including data centers, are large enough that connections must be batched and studied against transmission-system constraints.
- SF-ECON-LEAF-8D2A1131BCD0 Georgia Power 2025 IRP load growth from large-load customers - Georgia Power's IRP process identifies large-load growth, including data centers, as a planning driver for generation and grid-resource decisions in the Georgia/Carolinas load-growth belt.
- SF-ECON-LEAF-53DD356FFD70 Georgia PSC large-load and customer protection terms in Georgia Power IRP settlement - The Georgia PSC fact sheet frames large-load growth and ratepayer protections as part of the Georgia Power IRP settlement, supporting the cost-allocation/ratepayer-pressure lane.
- SF-ECON-LEAF-13D08FA1E3C6 cisa-niac-transformer-shortage-lead-time-stress - NIAC draft identifies a critical transformer shortage and multi-year procurement stress.
- SF-ECON-LEAF-2D6446203674 Texas directive for data centers to fund interconnection-related costs - The Texas directive links data-center interconnection to cost-allocation and ratepayer protection, instructing PUCT/ERCOT to prevent new large-load costs from being shifted to existing customers.
- SF-ECON-LEAF-31604FD300DB NOVEC data-center adjustment request for the 2026 PJM Load Forecast - PJM-hosted NOVEC material says projected data-center growth in Northern Virginia should be incorporated into the 2026 PJM Load Forecast, with 60+ data centers and about 1,300 MW load already served.
- SF-ECON-LEAF-D86B6A6A5825 EIA summary of PJM Dominion-zone peak-load growth driven by data centers - EIA reports PJM expects the Dominion zone to have the largest absolute summer peak-demand increase from 2026-2030, largely because of data-center load growth, citing PJM load forecast tables.
- SF-ECON-LEAF-B9486183FBB5 Virginia SCC-approved GS-5 large-load rate class for data centers - The SCC-approved GS-5 structure requires large-scale customers to pay minimum shares of contracted distribution/transmission and generation demand, explicitly aimed at insulating other ratepayers from data-center infrastructure costs.
- SF-ECON-LEAF-B80276CCFD4A Data-center operator filing links power availability and equipment delivery delays to expansion risk. - Data-center operator filing links power availability and equipment delivery delays to expansion risk.
- SF-ECON-LEAF-E561CE96A1D6 Supports treating narrative propagation as a measurable market-psychology mechanism rather than a purely impressionistic signal. - Supports treating narrative propagation as a measurable market-psychology mechanism rather than a purely impressionistic signal.
- SF-ECON-LEAF-D9125BD30499 Links a named shipping chokepoint to data-center project timing and cost risk in a primary company filing. - Links a named shipping chokepoint to data-center project timing and cost risk in a primary company filing.
- SF-ECON-LEAF-E7AD00CF08AA Provides current official maritime-risk context for the Red Sea transmission scenario. - Provides current official maritime-risk context for the Red Sea transmission scenario.
- SF-ECON-LEAF-5303581C067B Supports treating a boom as a conditional risk state rather than a deterministic crash forecast. - Supports treating a boom as a conditional risk state rather than a deterministic crash forecast.
- SF-ECON-LEAF-1CDF429528E2 Provides a theory in which real AI capital formation, elevated valuations, coordinated beliefs, and a later correction can coexist. - Provides a theory in which real AI capital formation, elevated valuations, coordinated beliefs, and a later correction can coexist.
- SF-ECON-LEAF-5CEE77A2C6A6 Supplies a dated example of how capital intensity, cash use, and financing availability can affect AI-infrastructure build-out risk. - Supplies a dated example of how capital intensity, cash use, and financing

availability can affect AI-infrastructure build-out risk.

- SF-ECON-LEAF-455E79778F9E micron-hbm-ai-data-center-demand - Micron investor materials identify HBM and data-center AI demand as a major memory growth driver.

- SF-ECON-LEAF-A0147824C8A6 micron-capex-hbm-data-center-allocation - Micron capital allocation materials emphasize investment toward data-center and high-value memory products.

- SF-ECON-LEAF-964C88978680 advanced-packaging-osat-ai-capacity - Industry materials point to advanced packaging and assembly capacity as a constraint in AI-chip production.

- SF-ECON-LEAF-3B3E3A348131 Power/cooling infrastructure supplier filing links strong demand, backlog, critical parts, logistics, and delivery pressure. - Power/cooling infrastructure supplier filing links strong demand, backlog, critical parts, logistics, and delivery pressure.

- SF-ECON-LEAF-D16B3C38A891 semi-equipment-capacity-cycle - SEMI market materials track semiconductor manufacturing equipment and fab capacity as a supply-cycle constraint.

- SF-ECON-LEAF-DF3792D87AC1 sk-hynix-hbm-ai-server-demand - SK hynix investor materials frame HBM as a core AI-server memory product with constrained high-end demand.

- SF-ECON-LEAF-30065FEBF040 samsung-hbm-memory-ai-demand - Samsung investor materials discuss HBM and high-performance memory demand tied to AI and server markets.

- SF-ECON-LEAF-885AB592F4FE tsmc-advanced-packaging-cowos-ai-bottleneck - TSMC investor materials identify advanced packaging capacity as part of AI accelerator supply expansion.

References And Evidence Receipts

1. **Forge Data Center Watch.** Data center bottleneck transmission and bubble mechanics packet. stock_intelligence_packets/data_center_reports/data_center_watch_latest.json

- Receipt SF-ECON-RCPT-9AA81AAAE52D | locator: locator not recorded | verification: verification status not recorded

2. **Forge Bubble Mechanics.** Psychological heat, capex-reality gap, survivor, and fragility scores. stock_intelligence_packets/data_center_reports/data_center_watch_latest.json

- Receipt SF-ECON-RCPT-291CACC66102 | locator: locator not recorded | verification: verification status not recorded

3. **Forge U.S. Power Crunch Watch.** Regional power and ratepayer-pressure packet. power_crunch_packets/power_crunch_us_latest.json

- Receipt SF-ECON-RCPT-1218F8AB2C7A | locator: locator not recorded | verification: verification status not recorded

4. **Local EIA AEO/EBA Archive Intake.** Local EIA projection and balancing-authority source context. power_crunch_packets/source_cache/eia_local_archive_evidence.json

- Receipt SF-ECON-RCPT-5A2EE3958768 | locator: locator not recorded | verification: verification status not recorded

5. **EIA Residential Retail Electricity Prices.** Measured state-level residential retail electricity-rate evidence. power_crunch_packets/source_cache/eia_residential_retail_prices.json

- Receipt SF-ECON-RCPT-EA8858A5D74E | locator: locator not recorded | verification: verification status not recorded

6. **Forge Utility/RTO Evidence Index.** Indexed utility, regulator, and grid-operator source candidates for regional cost-transmission evidence. power_crunch_packets/source_cache/utility_rto_evidence.json

- Receipt SF-ECON-RCPT-1392EE1D4222 | locator: locator not recorded | verification: verification status not recorded

7. **Forge Transformer And Grid-Equipment Supply Evidence.** Source-indexed transformer/switchgear lead-time, backlog, and supplier-filing evidence lane. power_crunch_packets/source_cache/transformer_supply_evidence.json

- Receipt SF-ECON-RCPT-8681EEA6FF9C | locator: locator not recorded | verification: verification status not recorded

8. **Forge Memory, HBM, And Chip-Production Evidence.** Source-indexed RAM/HBM, capex/capacity, and advanced-packaging evidence lane. power_crunch_packets/source_cache/memory_chip_supply_evidence.json

- Receipt SF-ECON-RCPT-EBDEBC6F2D21 | locator: locator not recorded | verification: verification status not recorded

9. **Forge Financial Deflation Lane.** Modeled market-structure lane for pricing-power fade and margin normalization. power_crunch_packets/power_crunch_us_latest.json

- Receipt SF-ECON-RCPT-7F9594B0A450 | locator: locator not recorded | verification: verification status not recorded

10. **Forge Cost Escalation And Economic Viability.** Modeled cost-stack sensitivity for energy, memory, grid, cooling, and construction pressure. power_crunch_packets/power_crunch_us_latest.json

- Receipt SF-ECON-RCPT-D7BFED6EF1C2 | locator: locator not recorded | verification: verification status not recorded

11. **Forge Geopolitical Shipping And Energy Lane.** Modeled chokepoint, logistics, insurance, and energy-risk transmission lane. power_crunch_packets/power_crunch_us_latest.json

- Receipt SF-ECON-RCPT-932D4B434E32 | locator: locator not recorded | verification: verification status not recorded

12. **Forge Shipping Risk Engine Refresh.** Latest Forge shipping-risk report linked into the power-crunch and paper evidence stack. power_crunch_packets/source_cache/shipping_risk_evidence.json

- Receipt SF-ECON-RCPT-D192DDB78460 | locator: locator not recorded | verification: verification status not recorded

13. **Forge Live Geopolitical Shipping/Energy Source Context.** Live public-source corroboration lanes for route-flow, freight, insurance, energy, and delivery-delay context. power_crunch_packets/source_cache/geopolitical_shipping_live_evidence.json

- Receipt SF-ECON-RCPT-DB73BF917E49 | locator: locator not recorded | verification: verification status not recorded

14. **Forge Geopolitical Causal Extraction.** Dated geopolitical event records and company delivery-delay records for shipping/energy scenario review. power_crunch_packets/source_cache/geopolitical_causal_extraction.json

- Receipt SF-ECON-RCPT-CDE751F87A9F | locator: locator not recorded | verification: verification status not recorded

15. **Forge Geopolitical Impact Calibration.** Source-linked freight, insurance, energy, and delivery-delay impact-lane calibration. power_crunch_packets/source_cache/geopolitical_impact_calibration.json

- Receipt SF-ECON-RCPT-48088EE445D5 | locator: locator not recorded | verification: verification status not recorded

16. **Forge Geopolitical Numeric Delta Replication.** Numeric-delta and proxy record layer for freight, insurance, energy, and delivery-delay lanes. power_crunch_packets/source_cache/geopolitical_numeric_delta_replication.json

- Receipt SF-ECON-RCPT-2A1AE77B08DB | locator: locator not recorded | verification: verification status not recorded

17. **Forge Data Center Bubble Evidence.** Measured market breadth and survivor-versus-fragility tracking artifact. stock_intelligence_packets/data_center_reports/data_center_bubble_evidence_latest.json

- Receipt SF-ECON-RCPT-6D0086048C87 | locator: locator not recorded | verification: verification status not recorded

18. **Forge Market Psychology Evidence Lane.** Narrative, credit/liquidity, and company guidance-language source context for the data-center bubble paper. stock_intelligence_packets/data_center_reports/market_psychology_evidence.json

- Receipt SF-ECON-RCPT-839B5F3EEB9C | locator: locator not recorded | verification: verification status not recorded

19. **DOE Large Power Transformer Resilience Report.** doe-lpt-resilience-2024-supply-chain-vulnerability. https://www.energy.gov/sites/default/files/2024-10/EXEC-2022-001242%20-%20Large%20Power%20Transformer%20Resilience%20Report%20signed%20by%20Secretary%20Granholm%20on%207-10-24.pdf

- Receipt SF-ECON-LEAF-EA8D22923777 | locator: record:doe-lpt-resilience-2024-supply-chain-vulnerability; date:2024-07-10 | verification: pending_human_review

- Exact supporting excerpt: DOE report frames large power transformer supply as a resilience and supply-chain vulnerability.

20. **DOE Office of Electricity Transformer Resilience and Advanced Components.** doe-trac-lpt-lead-time-lower-bound. https://www.energy.gov/oe/transformer-resilience-and-advanced-components-trac-program

- Receipt SF-ECON-LEAF-A912ED6FF535 | locator: TRAC Program page, Advanced Transformers section, accessed 2026-07-10 | verification: verified_primary_source

- Exact supporting excerpt: Large power transformers are custom engineered and can require lead times of one year or more.

21. **ERCOT trending topic on batch connection process for large electricity users.** ERCOT batch connection process for large electricity users. https://www.ercot.com/files/docs/2026/06/18/ERCOT-Trending-Topic-New-Batch-Connection-Process-for-Large-Electricity-Users.pdf

- Receipt SF-ECON-LEAF-5BEBDF51F0E2 | locator: record:UTILITY-RTO-TX-ERCOT-LARGE-LOAD-BATCH-2026-001; date:2026-06-18 | verification: pending_human_review

- Exact supporting excerpt: ERCOT is tracking more than 438,000 MW of new connection requests from large users like data centers.

22. **Georgia Power 2025 Integrated Resource Plan.** Georgia Power 2025 IRP load growth from large-load customers. https://www.georgiapower.com/company/energy-industry/energy-sources/integrated-resource-plan.html

- Receipt SF-ECON-LEAF-8D2A1131BCD0 | locator: record:UTILITY-RTO-GA-GPC-IRP-DATA-CENTER-LOAD-2025-001; date:2025-01-31 | verification: pending_human_review

- Exact supporting excerpt: Georgia Power expects load growth from new large-load customers including data centers and manufacturing.

23. **Georgia Public Service Commission fact sheet on 2025 Georgia Power IRP settlement.** Georgia PSC large-load and customer protection terms in Georgia Power IRP settlement. https://psc.ga.gov/site/assets/files/8378/fact_sheet_-_2025_georgia_power_integrated_resource_plan_stipulation_6-17-25.pdf

- Receipt SF-ECON-LEAF-53DD356FFD70 | locator: record:UTILITY-RTO-GA-PSC-LARGE-LOAD-PROTECTION-2025-002; date:2025-06-17 | verification: pending_human_review

- Exact supporting excerpt: Customer protections are established for large load customers to ensure costs are allocated fairly.

24. **NIAC Draft Report on Critical Shortage of Power Transformers.** cisa-niac-transformer-shortage-lead-time-stress. https://www.cisa.gov/sites/default/files/2024-06/DRAFT_NIAC_Addressing%20the%20Critical%20Shortage%20of%20Power%20Transformers%20to%20Ensure%20Reliability%20of%20the%20U.S.%20Grid_Report_06052024_508c.pdf

- Receipt SF-ECON-LEAF-13D08FA1E3C6 | locator: record:cisa-niac-transformer-shortage-lead-time-stress; date:2024-06-05 | verification: pending_human_review

- Exact supporting excerpt: NIAC draft identifies a critical transformer shortage and multi-year procurement stress.

25. **Office of the Texas Governor directive to PUCT and ERCOT.** Texas directive for data centers to fund interconnection-related costs. https://gov.texas.gov/news/post/governor-abbott-directs-puc-and-ercot-to-shield-texans-from-data-center-infrastructure-costs

- Receipt SF-ECON-LEAF-2D6446203674 | locator: Official press release, June 10, 2026, ratepayer-protection and data-center infrastructure directives | verification: verified_primary_source

- Exact supporting excerpt: PUC must require data centers to fully fund the costs of interconnecting to the ERCOT power grid.

26. **PJM Load Analysis Subcommittee / NOVEC Data Center Load Forecast.** NOVEC data-center adjustment request for the 2026 PJM Load Forecast. https://www.pjm.com/-/media/DotCom/committees-groups/subcommittees/las/2025/20250916/20250916-item-04b---novec-large-load-request.pdf

- Receipt SF-ECON-LEAF-31604FD300DB | locator: PDF pp.1-2 and 13-14; Key Takeaways, About NOVEC, and 2026 load-forecast adjustment request | verification: verified_primary_source
- Exact supporting excerpt: PJM should incorporate NOVEC data-center growth as a 2026 load-forecast adjustment.

27. **U.S. EIA Today in Energy using 2026 PJM Load Forecast Report tables.** EIA summary of PJM Dominion-zone peak-load growth driven by data centers. https://www.eia.gov/todayinenergy/detail.php?id=67664

- Receipt SF-ECON-LEAF-D86B6A6A5825 | locator: record:UTILITY-RTO-NOVA-EIA-PJM-DOMINION-2026-LF-002; date:2026-05-05 | verification: pending_human_review
- Exact supporting excerpt: Dominion zone has the largest absolute summer peak-demand increase, largely from data-center load.

28. **Virginia SCC action summarized by American Action Forum with SCC citation.** Virginia SCC-approved GS-5 large-load rate class for data centers. https://www.scc.virginia.gov/about-the-scc/newsreleases/release/scc-issues-order-on-dev-biennial-review-2025/scc-rules-in-dev-biennial-review-case.html

- Receipt SF-ECON-LEAF-B9486183FBB5 | locator: record:UTILITY-RTO-NOVA-SCC-GS5-RATE-CLASS-003; date:2026-01-08 | verification: pending_human_review
- Exact supporting excerpt: The SCC approved a new electricity rate for large-scale customers, notably AI data centers.

29. **SEC EDGAR.** Data-center operator filing links power availability and equipment delivery delays to expansion risk.. https://www.sec.gov/Archives/edgar/data/1101239/000110123925000062/eqix-20250930.htm

- Receipt SF-ECON-LEAF-B80276CCFD4A | locator: 2025 Q3 Form 10-Q, Item 2, pp.40-41, power limitations and global supply-chain challenges | verification: verified_primary_source
- Exact supporting excerpt: Equinix disclosed that power limitations could constrain data-center expansion and that global supply-chain challenges could delay data-center equipment delivery.

30. **NBER Working Paper 30195: Crash Narratives.** Supports treating narrative propagation as a measurable market-psychology mechanism rather than a purely impressionistic signal.. https://www.nber.org/papers/w30195

- Receipt SF-ECON-LEAF-E561CE96A1D6 | locator: NBER Working Paper 30195, abstract, lines 141-142; revised November 2024 | verification: verified_primary_source
- Exact supporting excerpt: Crash narratives propagate broadly once they appear in news articles, and they predict market volatility.

31. **Equinix 2025 Q3 Form 10-Q.** Links a named shipping chokepoint to data-center project timing and cost risk in a primary company filing.. https://www.sec.gov/Archives/edgar/data/1101239/000110123925000062/eqix-20250930.htm

- Receipt SF-ECON-LEAF-D9125BD30499 | locator: Equinix 2025 Q3 Form 10-Q, Item 1A, p.75, lines 3079-3081 | verification: verified_primary_source
- Exact supporting excerpt: Equinix states that Red Sea attacks disrupt shipping routes; alternative routes can add transit time, delay deliveries, and increase fuel costs, affecting planned expansion-project costs.

32. **U.S. Maritime Administration Advisory 2026-006.** Provides current official maritime-risk context for the Red Sea transmission scenario.. https://www.maritime.dot.gov/msci/2026-006-red-sea-bab-el-mandeb-strait-gulf-aden-arabian-sea-and-somali-basin-houthi-attacks

- Receipt SF-ECON-LEAF-E7AD00CF08AA | locator: U.S. Maritime Advisory 2026-006, Issue section, lines 204-208; effective 2026-03-26 to 2026-09-22 | verification: verified_primary_source

- Exact supporting excerpt: MARAD identifies a continuing threat to commercial vessels in the Red Sea, Bab el Mandeb Strait, and Gulf of Aden, with more than 100 separate Houthi attacks on commercial vessels from November 2023 to October 2025.

33. **NBER Working Paper 21693: Bubble Investing: Learning from History.** Supports treating a boom as a conditional risk state rather than a deterministic crash forecast.. https://www.nber.org/papers/w21693

- Receipt SF-ECON-LEAF-5303581C067B | locator: NBER Working Paper 21693, abstract, lines 141-142; revised January 2016 | verification: verified_primary_source

- Exact supporting excerpt: The probability of a crash conditional on a boom is only slightly higher than the unconditional probability; not all booms are bad.

34. **NBER Working Paper 34722: Speculative Growth and the AI Bubble.** Provides a theory in which real AI capital formation, elevated valuations, coordinated beliefs, and a later correction can coexist.. https://www.nber.org/papers/w34722

- Receipt SF-ECON-LEAF-1CDF429528E2 | locator: NBER Working Paper 34722, abstract, lines 139-140; issued January 2026, revised June 2026 | verification: verified_primary_source

- Exact supporting excerpt: AI technology can generate speculative-growth equilibria; elevated valuations can sustain capital accumulation, while a loss of confidence can precipitate a self-fulfilling crash and reversal.

35. **CoreWeave 2025 Form 10-K.** Supplies a dated example of how capital intensity, cash use, and financing availability can affect AI-infrastructure build-out risk.. https://www.sec.gov/Archives/edgar/data/1769628/000176962826000104/crwv-20251231.htm

- Receipt SF-ECON-LEAF-5CEE77A2C6A6 | locator: CoreWeave 2025 Form 10-K, Liquidity and Capital Resources, pp.75-76, lines 2374-2377 and 2401-2412 | verification: verified_primary_source

- Exact supporting excerpt: CoreWeave states that future investments may require significant debt and/or equity financing, while reporting substantial infrastructure capital investment and negative investing cash flows.

36. **Micron Investor Relations.** micron-hbm-ai-data-center-demand. https://investors.micron.com/

- Receipt SF-ECON-LEAF-455E79778F9E | locator: record:micron-hbm-ai-data-center-demand; date:2025 | verification: weak_context_needs_replacement

- Exact supporting excerpt: Micron investor materials identify HBM and data-center AI demand as a major memory growth driver.

37. **Micron Investor Relations.** micron-capex-hbm-data-center-allocation. https://investors.micron.com/

- Receipt SF-ECON-LEAF-A0147824C8A6 | locator: record:micron-capex-hbm-data-center-allocation; date:2025 | verification: weak_context_needs_replacement

- Exact supporting excerpt: Micron capital allocation materials emphasize investment toward data-center and high-value memory products.

38. **OSAT And Advanced Packaging Supplier Materials.** advanced-packaging-osat-ai-capacity. https://www.semi.org/en/news-media-press

- Receipt SF-ECON-LEAF-964C88978680 | locator: record:advanced-packaging-osat-ai-capacity; date:2025 | verification: weak_context_needs_replacement

- Exact supporting excerpt: Industry materials point to advanced packaging and assembly capacity as a constraint in AI-chip production.

39. **SEC EDGAR.** Power/cooling infrastructure supplier filing links strong demand, backlog, critical parts, logistics, and delivery pressure.. https://www.sec.gov/Archives/edgar/data/1674101/000167410126000008/vrt-20251231.htm

- Receipt SF-ECON-LEAF-3B3E3A348131 | locator: 2025 Form 10-K, Item 1 p.9 Raw Materials and Supplies; Item 1 p.6 Backlog | verification: verified_primary_source

- Exact supporting excerpt: Vertiv disclosed that critical-part shortages, premium freight, and logistical issues may create production and delivery challenges; it also reported backlog growth tied to demand ahead of fulfillment capacity.

40. **SEMI Semiconductor Manufacturing And Equipment Market Data.** semi-equipment-capacity-cycle. https://www.semi.org/en/news-media-press

- Receipt SF-ECON-LEAF-D16B3C38A891 | locator: record:semi-equipment-capacity-cycle; date:2025 | verification: weak_context_needs_replacement
- Exact supporting excerpt: SEMI market materials track semiconductor manufacturing equipment and fab capacity as a supply-cycle constraint.

41. **SK hynix Investor Relations.** sk-hynix-hbm-ai-server-demand. https://www.skhynix.com/ir/UI-FR-IR01

- Receipt SF-ECON-LEAF-DF3792D87AC1 | locator: record:sk-hynix-hbm-ai-server-demand; date:2025 | verification: pending_human_review
- Exact supporting excerpt: SK hynix investor materials frame HBM as a core AI-server memory product with constrained high-end demand.

42. **Samsung Electronics Investor Relations.** samsung-hbm-memory-ai-demand. https://www.samsung.com/global/ir/

- Receipt SF-ECON-LEAF-30065FEBF040 | locator: record:samsung-hbm-memory-ai-demand; date:2025 | verification: weak_context_needs_replacement
- Exact supporting excerpt: Samsung investor materials discuss HBM and high-performance memory demand tied to AI and server markets.

43. **TSMC Investor Relations.** tsmc-advanced-packaging-cowos-ai-bottleneck. https://investor.tsmc.com/english

- Receipt SF-ECON-LEAF-885AB592F4FE | locator: record:tsmc-advanced-packaging-cowos-ai-bottleneck; date:2025 | verification: pending_human_review
- Exact supporting excerpt: TSMC investor materials identify advanced packaging capacity as part of AI accelerator supply expansion.

Evidence Standards

- No investment recommendation language.
- No deterministic bubble-burst date without a pre-registered trigger model.
- No national household-cost claim without regional evidence.
- Public presentation must distinguish checked facts, historical model results and the Forge's interpretations.
- Validated-forecast language remains prohibited until scenario outcomes mature and are scored.